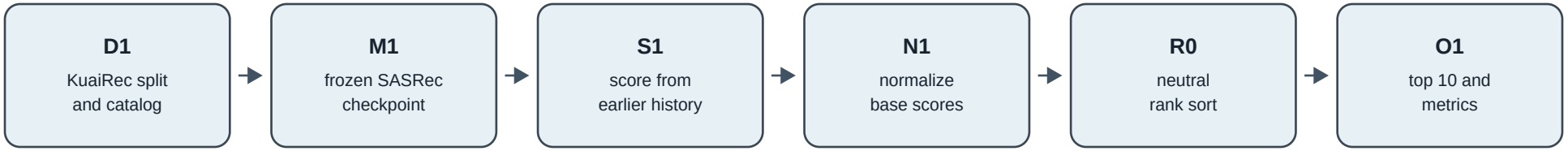
COMP 9500 pipeline and LLM boundary

Code mapping and precise limits: PIPELINE-AND-CODE-WALKTHROUGH.md | FP bec8442 | 27 Oct 2026

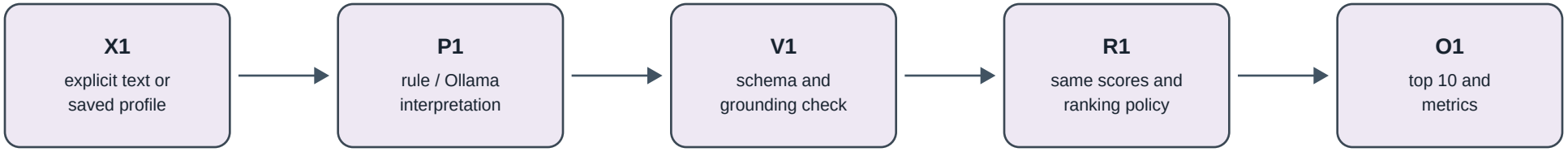
1 Uncontrolled history-only reference (not Condition A after feedback)



2 A/B item feedback (one item; B adds display-only explanations)

3 Independently established human intent: conceptual reference, not a live backend

4 Implemented control language path: rule or local Llama parsing enters at P1



E1 reads O1. U1 displays O1; EX1 can use earlier cached Llama fact selection or a template, without reranking.
A rule-mode feed change proves shared control enforcement, not unique LLM benefit or human satisfaction.

Data, score and comparison paths

Follow the block IDs on page 1; these are implementation paths, not independent experiments.

D1 KuaiRec input and split

feedctrl/data.py:91-194 builds the processed catalog, user histories and split metadata. Training, request-history and final-test windows are separate. Synthetic E1 requests use earlier history; the held-out outcomes do not become real user intent. Check the saved manifest before reproducing an evaluation.

M1 and S1 Frozen model and base scores

feedctrl/model.py:26-88 embeds item IDs and positions, passes histories through two causal attention/pointwise blocks (hidden 32, one head, last 50 items), then scores each candidate by dot product with the final history state (model.py:203-223). Saved seed-42 trained model was used for the worked example. Diagnostic scoring is a separately labelled fallback.

How M1 was trained

_training_examples at model.py:89-103 uses train transitions only. train_model at 106-190 fits one uniform negative per valid step with positive/negative softplus loss, Adam and gradient clipping, then stores checkpoint and metadata. Saved seed-42 metadata records 20 fixed epochs, batch 64, learning rate 0.001; seeds 42-44 are used in the study. No request-history or test item enters gradients; this is an exploratory backbone, not a convergence claim.

N1 and R0 Untouched history-only reference

controls.normalize_scores at feedctrl/controls.py:297-305 normalizes base scores. Empty command and profile produce a neutral sort through rerank at lines 326-339. R0 is the history-only calculation; current Condition A is a different comparator once item feedback is applied.

A1 Manual UI feedback proxy

The A/B path adds 0.25 to one liked item and excludes a disliked item from the returned ranking (api.py:150-157). /api/feedback at 295-305 stores the latest direction; clear removes that stored choice. evaluation.item_feedback_ranking at 75-84 is the offline proxy. KuaiRec has no observed thumbs labels.

H0 Manual known-intent example versus study

The packet already demonstrates explicit category_1 boost and category_28 mute with known specified actions; they can enter V1 and the same R1 policy. H0 is not a separate live human-parser backend. The completed student worksheet, 99/100 comparison and exposed-practice 73/100 versus 18/100 parser check are descriptive, not a fresh held-out human ranking evaluation.

From language to a changed feed

Only P1 is the candidate language-understanding advantage; the recommender and enforcement are shared.

X1 and P1 Text, profile, rule or Ollama

api.py:243-293 accepts control text and editable profile text. controls.parse_control at 243-254 and parse_profile at 257-294 select rule or local Ollama. One free-text control requires exactly one explicit category_N ID; a profile can contain multiple allowed explicit IDs. Named topics are not inferred. Parsed JSON is a proposed action, not a feed change by itself.

V1 Validate before state changes

controls._validate_control at 104-119 checks keys, operation, fixed strength and allowed category. Textual grounding to exactly one explicit ID occurs in parse_control at 250-253; profile grounding occurs in parse_profile at 291-292. Invalid or ambiguous language fails or clarifies. Ollama error returns HTTP 502 and preserves prior state, with no silent rule fallback.

R1 and O1 Shared policy, rank and response

controls.apply_to_profile at 308-323 gives an explicit same-category command precedence. rerank at 326-339 adds category boost weights and hard-excludes muted-category items before sorting. The top 10, scores, preference counts and notices are returned through feedctrl/api.py. The rule and LLM paths use the same scoring and policy after validation.

A-E conditions and explanation boundary

A: one-item feedback. B: A plus display-only explanations, so order stays equal. C: parsed free-text category control. D: saved explicit profile preference. E: profile plus explicit control; the later same-category command wins rather than double-counting. EX1 in feedctrl/explanations.py:20-86 selects constrained saved facts or a labelled template; an earlier cached Llama selection may be present, but display does not rerank and human response was not measured.

One executed rule-mode example, not a new science result

With saved processed data, user 14 and seed-42 model: neutral top three 8958, 7383, 1116. Like item 1116 makes it first in A and B. Explicit category_1 boost yields nine of ten target items in matched C/D/E; muting category_1 in E removes them. The receipt and reproducer are in the companion packet. No fresh LLM call or browser action is implied.

Current language limit

The rule and Ollama prompts map less/fewer to HARD mute because this UI has no quantitative soft decrease. category_N identifiers are opaque and must be explicit. A successful parse therefore demonstrates a narrow grounded command, not universal free-language understanding or satisfaction.

Visible UI route

web/src/main.jsx gets session and status, selects user and A-E condition, calls GET /api/feed, POST /api/feedback or /api/control or /api/profile/text, then refreshes feed/notice/count. GET /api/compare requests A-E views; POST /api/reset clears this user's controls and item feedback. Read data/model/parser badges before describing a live result.

Evaluation and what the result means

Use the numbers with their protocol and denominator; avoid treating an exposure metric as human satisfaction.

E1 Metric calculation

evaluation.ranking_metrics at 51-72 evaluates fixed top K=10: TCP is target-category item count divided by ten for positive requests; TCER is the same exposure fraction for mutes. Fill reports how many slots remain occupied. HR@1 and NDCG@10 compare against historical next-item relevance. User-level paired summaries average three fixed seeds; request-seed rows are not independent users.

Hand-worked top-ten count

For an independently specified boost of category_1, six matching items among ten gives TCP=6/10=0.6 and fill=10/10=1. For a category_28 mute, two matches among ten gives TCER=2/10=0.2, a failure of full removal. Count the REQUESTED category even if a parser returns a different category; otherwise parsing errors disappear from the metric. E2's induced-movement/success measures use a separate reference and protocol.

Frozen Experiment 1

Canonical rule and Llama rankings are identical, so E1 feed changes show shared category-policy enforcement, not an LLM-specific gain. TCP rose 11.6% to 79.2% and TCER fell 6.1% to zero under their assigned requests; boost NDCG@10 fell 45.7%. Broader synthetic parser accuracy was Llama 79/100 versus rules 81/100, below the proposal's 85% target.

Revised AI-reference Experiment 2 and separate student check

The completed fresh AI-reference E2 reports whole-command agreement Llama 76/100 versus rules 11/100 and induced success 20.145% versus item-feedback 3.815%; NDCG was 0.02580 versus 0.03184 and conditional success was zero. References are model-generated. The separately dated student-label exposed-practice parser result is 73/100 versus 18/100; it is not the same fresh evaluation.

Validity boundary and professor's decision

Offline TCP/TCER can verify exposure under an assigned category command. They do not establish that a person wanted that command, found the result useful, or remained satisfied. The original archived human-reference route was not executed; the revised AI-reference study is complete. A new independently specified human-intent protocol is a proposed next study only if the supervisor requests it.